

**FIRST, A LOOK AT
ERRORS**

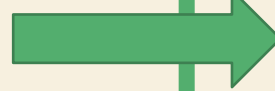
QUESTION 1: WHAT'S WRONG WITH THIS CODE?

```
cin >> enter dep(A, B, C, D)
```

```
enter salary dep = toupper(dep);
```

ANSWER 1: WHAT'S WRONG WITH THIS CODE?

`cin >> enterdep(A, B, C, D)`



```
cout << "Enter a department: ";  
cin >> dep;
```

`enter salary dep = toupper(dep);`



```
dep = toupper(dep);
```

QUESTION 2: WHAT'S WRONG WITH THIS CODE?

```
cout << "Enter a department; ";
cin >> dep;

switch (dept)
{
case 'A':
case 'B':
    raise = salary * .02;
    cout >> "Your raise is $" << raise << endl;
    break;
case 'C':
    raise = salary = 0.15;
    cout >> "Your raise is $" << raise << endl;
    break;
case 'D':
    raise = salary = 0.3;
    cout >> "Your raise is $" << raise << endl;
    break
default:
    cout << "You have chose an incorrect department."
} //endswitch
```

ANSWER 2: WHAT'S WRONG WITH THIS CODE?

```
cout << "Enter a department; ";
cin >> dep;

switch (dept)
{
case 'A':
case 'B':
    raise = salary * .02;
    cout >> "Your raise is $" << raise << endl;
    break;

case 'C':
    raise = salary = 0.15;
    cout >> "your raise is $" << raise << endl;
    break;

case 'D':
    raise = salary = 0.3;
    cout >> "your raise is $" << raise << endl;
    break;

default:
    cout << "You have chose an incorrect
department."
} //endswitch
```

```
cout << "Enter a department; ";
cin >> dep;
dep = toupper(dep);
switch (dept)
{
    case 'A':
    case 'B':
        raise = salary * .02;
        cout << "Your raise is $ " << raise << endl;
        break;
```

CHAPTER 7

THE BEGINNING

THE REPETITION STRUCTURE



KEY WORDS TO KNOW

- Repetition structure
- Loop
- Looping condition
- Loop exit condition
- Loop body
- Pretest loop
- Sentinel values
- Priming read
- Update read
- While statement
- Endless loop
- Infinite loop
- Counter
- Accumulator
- Updating
- Incrementing
- Decrementing
- Counter-controlled loops
- For statement

LOOP EXAMPLE 1

```
cout << "Enter a positive or negative integer (enter 0 to end):  ";
```

```
cin >> number;    //priming read for the loop
```

```
while (number != 0)    //zero is the sentinel value
                        // (number !=0) is the looping condition
{
    if (number > 0)
        positive += 1;
    else
        negative += 1;
    //end if

    cout << "Enter another positive or negative integer (enter 0 to end): ";
    cin >> number; //update read for the loop
} //end while
```

LOOP EXAMPLE 2

```
int number = 1; //in this example, this is the priming read for the loop
```

```
while (number < 5) //looping condition
```

```
{ //start of loop body
```

```
    cout << number << endl;
```

```
    number += 1; //update read
```

```
} //end while
```

LOOP EXAMPLE 3

```
int age = 0;
cout << "Enter an age: ";
cin>> age;    //priming read

while (age > 0)    //loop condition; //pretest loop
{
    cout << "You entered age:" << age << endl;
    cout << "Enter an age: ";    //update read
    cin>> age;    //part of update read
}
```

LOOP EXAMPLE 4

```
char anotherSale = ' '; //this will be used as a flag or switch
double sales = 0.0;
cout << "Enter a sales amount? (Y/N) ";
cin >> anotherSale;
while (toupper(anotherSale) == 'Y') //make the flag uppercase for comparisons
{
    cout << "Enter sales amount ";
    cin >> sales;
    cout << "You entered " << sales << endl;
    cout << "Enter a sales amount? (Y/N) "; //this is the update read
    cin >> anotherSale; //part of the update read
}
```

ENDLESS LOOP EXAMPLE

```
int age = 0;
cout << "Enter an age: ";
cin >> age;    //priming read

while (age > 0)    //loop condition; //pretest loop
{
    cout << "You entered age:" << age << endl;
}
```